

BattingEdge V9.5: AI Model Performance Report

Algorithm: Temporal Convolutional Network (TCN) | Architecture: Dilated Causal Convolutions

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1. Executive Summary

The **BattingEdge V9.5** Temporal Convolutional Network (TCN) experiment aimed to leverage dilated convolutions for capturing long-range dependencies in cricket shot sequences. The model achieved an **Overall Test Accuracy of 61.90%**, which is significantly below the project's minimum viability threshold (83%).

The model exhibited catastrophic confusion between distinct shot types, particularly favoring the "Cut Shot" class as a default prediction for ambiguous movements. This failure highlights the TCN architecture's struggle to model fine-grained skeletal dynamics with limited data compared to recurrent architectures (LSTM/GRU).

2. Methodology & Architecture

2.1 Model Design

- **Core:** 1D Dilated Causal Convolutions. The network used dilation factors of [1, 2, 4, 8, 16] to exponentially increase the Receptive Field, theoretically allowing the top layer to see the entire 50-frame sequence without recurrent loops.
- **Residual Connections:** Skip connections were implemented to facilitate gradient flow and prevent signal degradation in deeper layers.
- **Input:** The standard 50-frame sequence of 107 hybrid features (99 pose + 8 angles).

2.2 Dataset Split

- **Training Set:** 3,007 samples
 - **Validation Set:** 388 samples
 - **Test Set:** 378 samples (Unseen data)
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3. Performance Metrics

3.1 Global Metrics

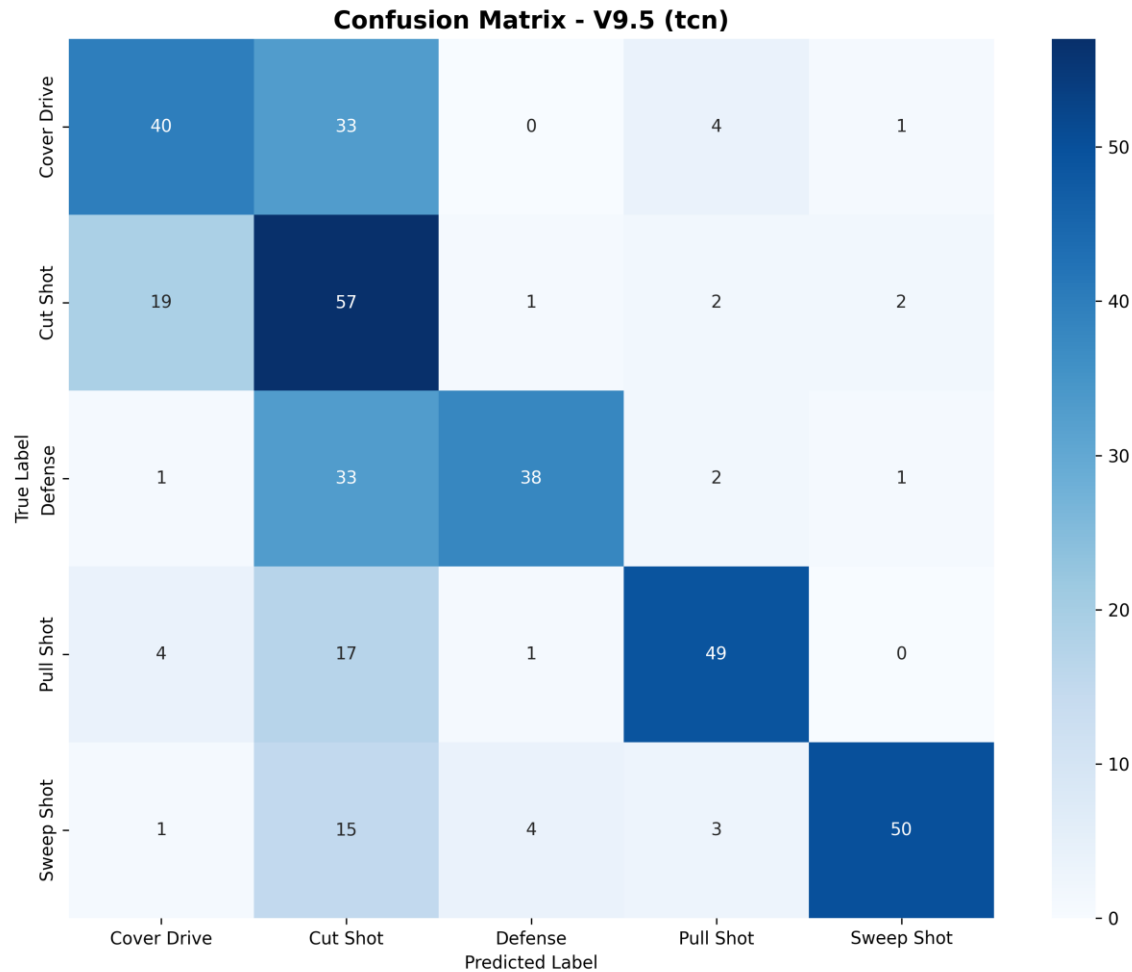
Metric	Score	Assessment
Accuracy	61.90%	Critical Failure. Far below the LSTM benchmark (91.8%).
Macro F1-Score	0.64	Indicates poor generalization across classes.
Convergence	Slow	The loss curve plateaued early (around 1.0), indicating the model failed to learn complex patterns.

3.2 Per-Class Analysis

The model effectively collapsed into a "Cut Shot Detector," misclassifying nearly every other shot type as a Cut.

Shot Type	Precision	Recall	Accuracy	Assessment
Cut Shot	0.37	0.70	70.4%	Over-Predicted. High recall but terrible precision (lots of false positives).
Sweep Shot	0.93	0.69	68.5%	Passable. Only worked because the pose is extremely distinct.
Pull Shot	0.82	0.69	69.0%	Weak. Confused with Cut shots constantly.
Cover Drive	0.62	0.51	51.3%	Broken. A coin flip is nearly as accurate.
Defense	0.86	0.51	50.7%	Broken. Half of all defensive shots were missed.

4. Error Analysis (Confusion Matrix)



The Confusion Matrix reveals a systemic bias:

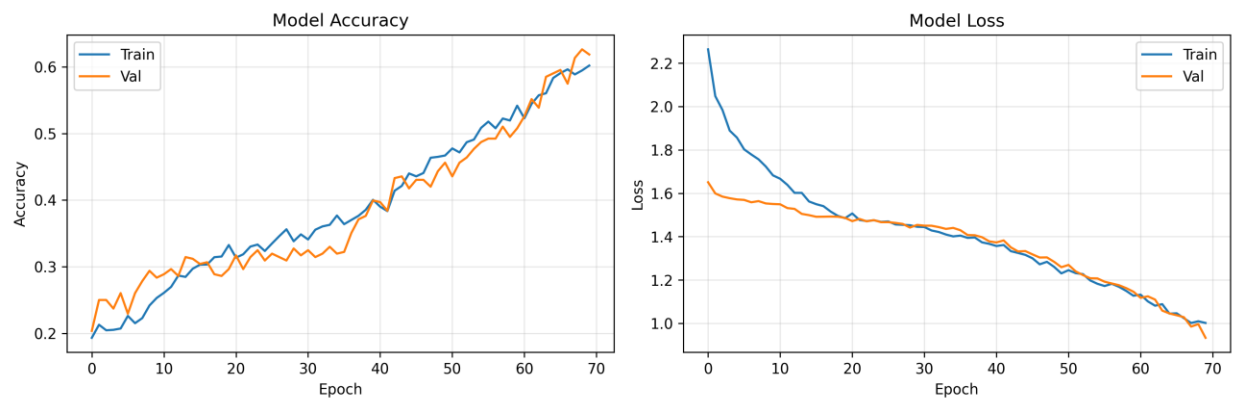
4.1 The "Cut Shot" Trap

- **Observation:** The model predicted "Cut Shot" **155 times**, but there were only 81 actual Cut Shots in the test set.
- **Analysis:**
 - **Cover Drive -> Cut Shot (33 cases):** 42% of drives were called cuts.
 - **Defense -> Cut Shot (33 cases):** 44% of defenses were called cuts.
 - **Pull Shot -> Cut Shot (17 cases):** 24% of pulls were called cuts.
 - **Sweep Shot -> Cut Shot (15 cases):** 20% of sweeps were called cuts.

4.2 Root Cause Hypothesis

The TCN filters likely latched onto a simple, high-frequency feature—such as the **lateral movement of the wrists**—which is present in Cut shots. Because the TCN processes frames in parallel (convolution) rather than sequentially (recurrence), it failed to learn the *context* (e.g., "front foot movement must happen *before* the wrist movement"). Without this temporal context, any shot with wrist movement was classified as a "Cut."

5. Conclusion & Recommendation



Verdict

The TCN architecture is rejected for this application.

While TCNs are powerful for audio waveform or stock price prediction, they struggled here to capture the subtle biomechanical sequencing required to distinguish similar cricket shots (e.g., Defense vs. Drive) on a dataset of this size.

Report generated by BattingEdge AI Analysis Team.